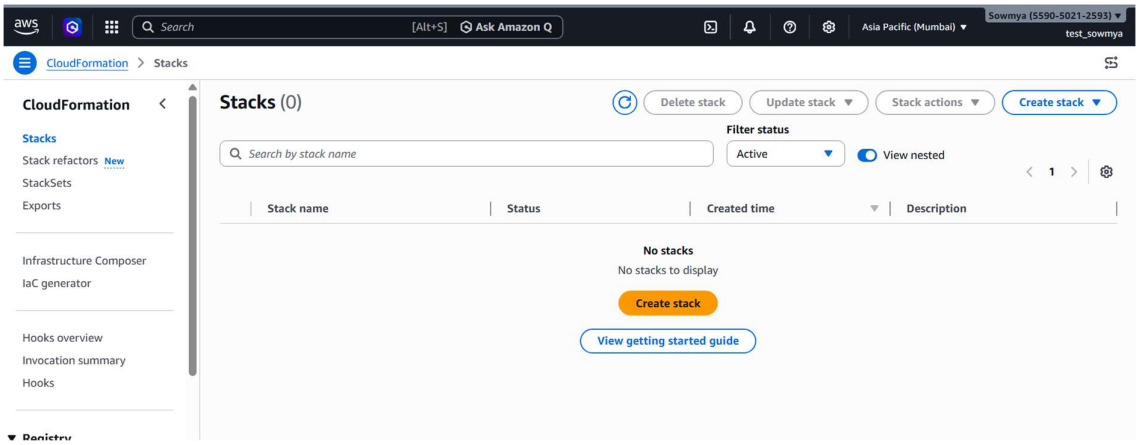
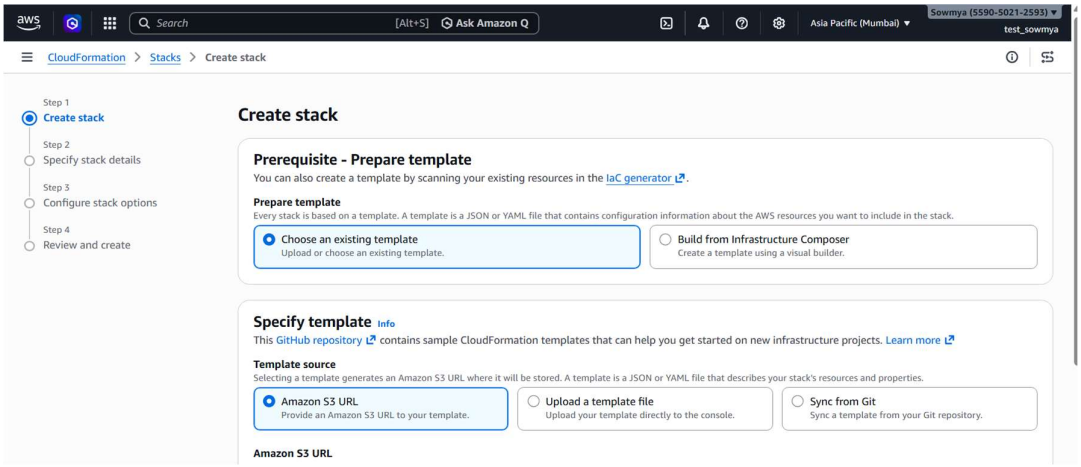
**IMPLEMENTATION:-CLOUD FORMATION Templates**

**Step 1:** Login to AWS Console with IAM user credentials and search for cloud formation

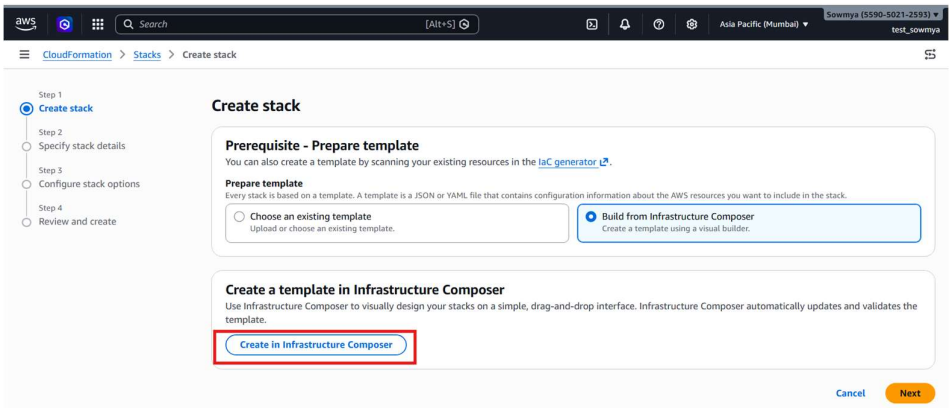
**Step 2**: You will see as create stack as in below image. Click on it



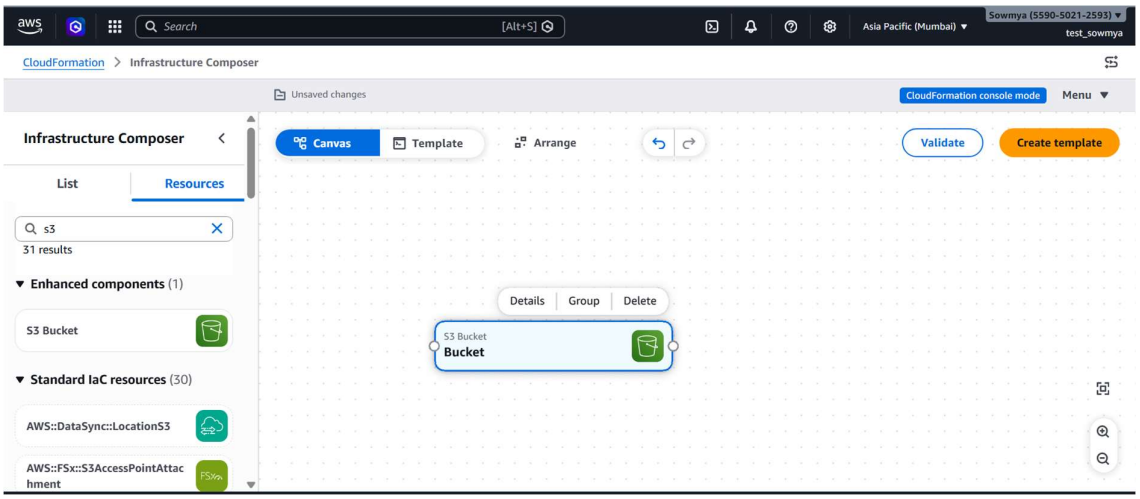

**Step 3**: You will be able to see 2 options to create template. You can either use one of them.



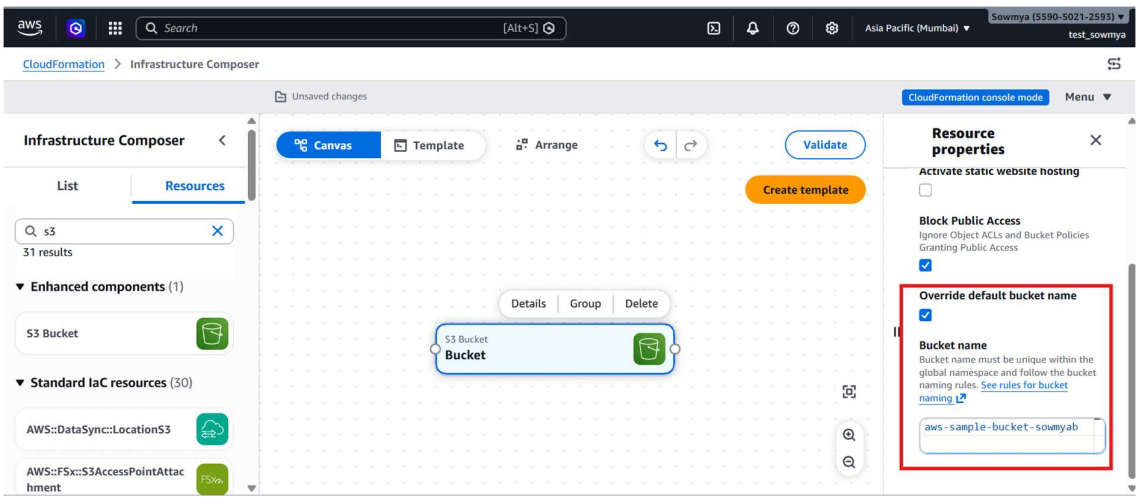

**Step 4**: Let's try to create template with build from infrastructure composer. Click on create in infrastructure composer.



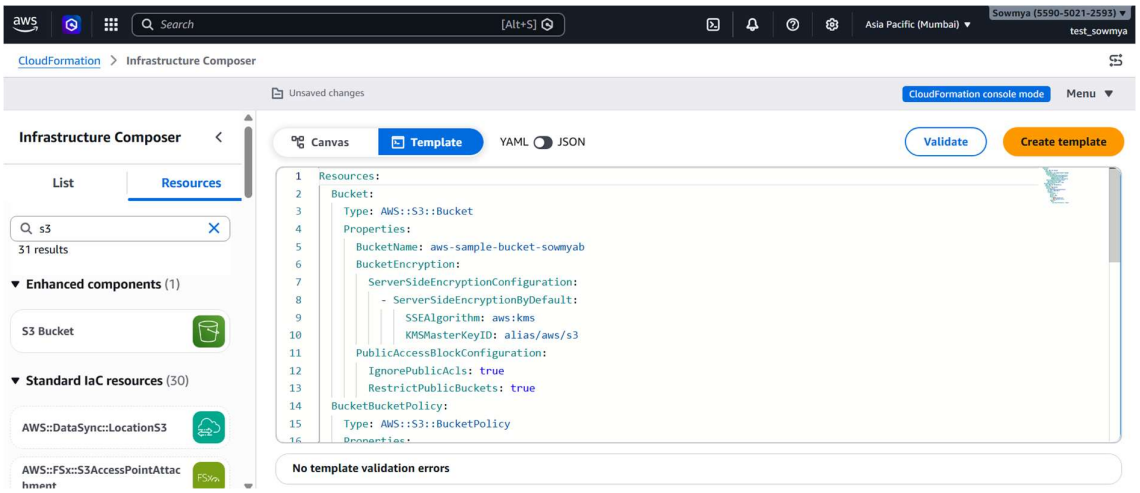
**Step 5**: Drag the resource from left side that you want to create the template for and keep it on the blank canvas, like in below image.



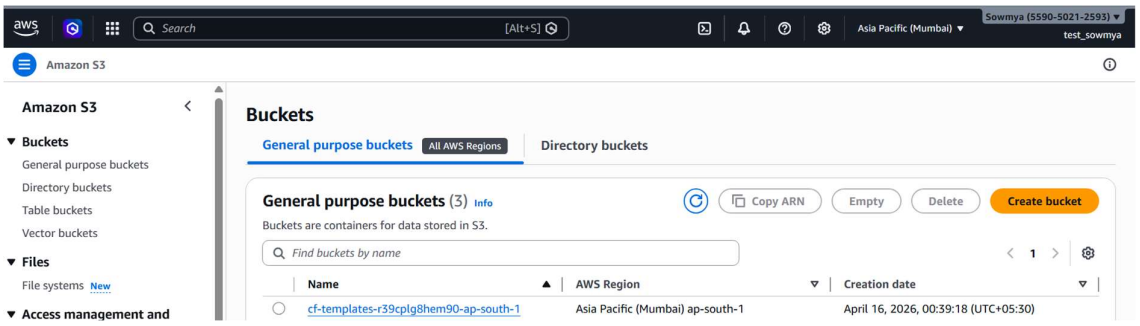

**Step 6**: Click on the bucket and resource properties will open in right-side corner. There tick the override default bucket name and give the bucket name of your choice. Click on save



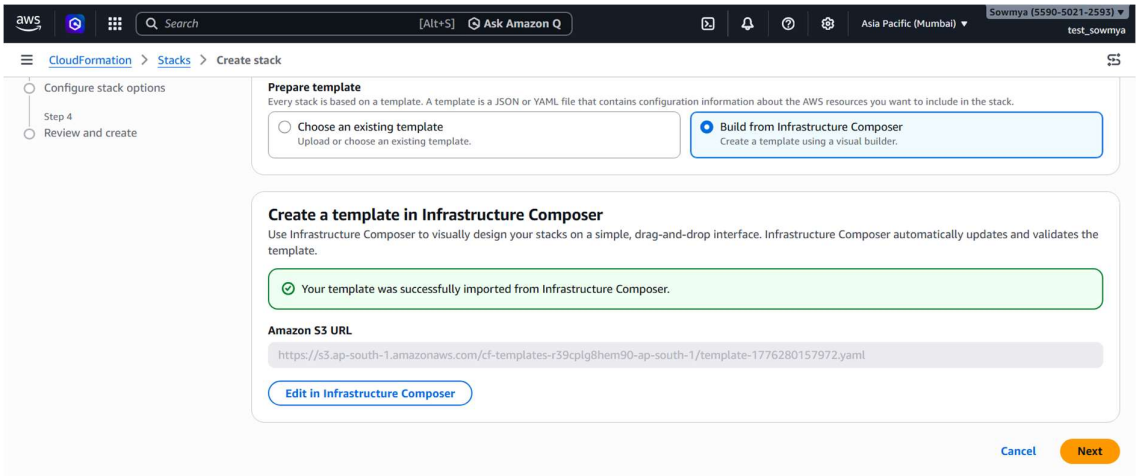

**Step 7**: In the top you have Template button. Click on it and check the template. If you want add additional properties you can add it in the template. If everything looks good, click on validate on top.

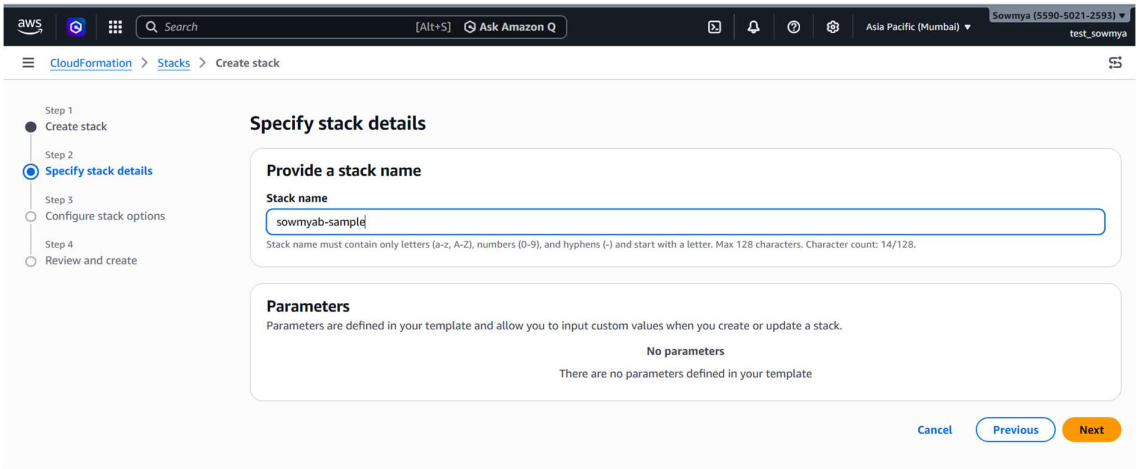



**Step 8**: Click on Create template. This will generate a CloudFormation template, which can be used to create a stack and provision the S3 bucket.

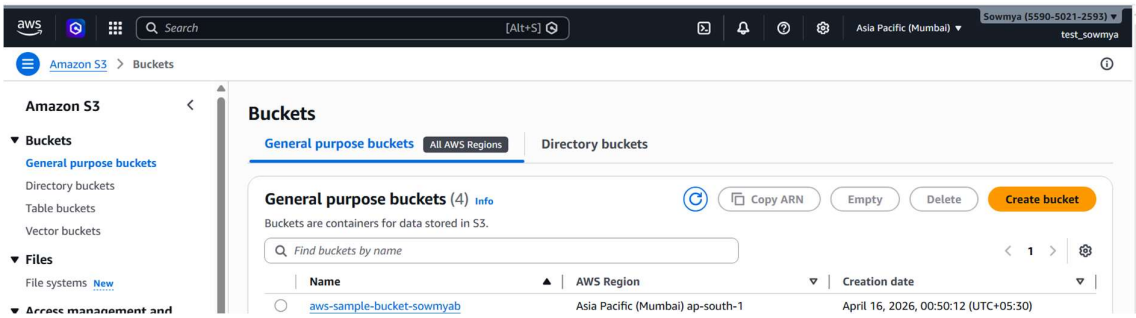



**Step 9:** Go back to your stack creation tab ,click on next and give stack name and then click on next review everything. Once done click on submit.

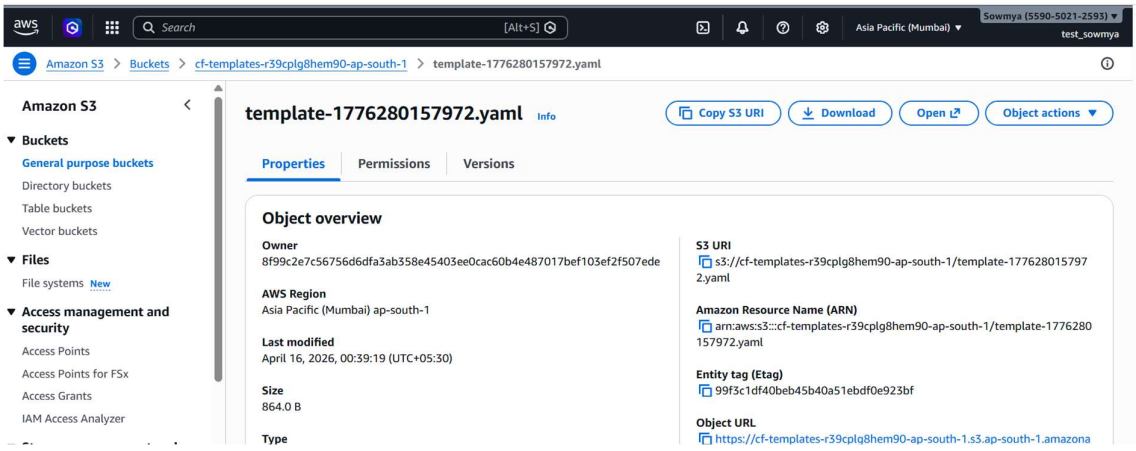





**Step 10**: Once the creation is complete, go back to s3 and check if bucket is created or not.



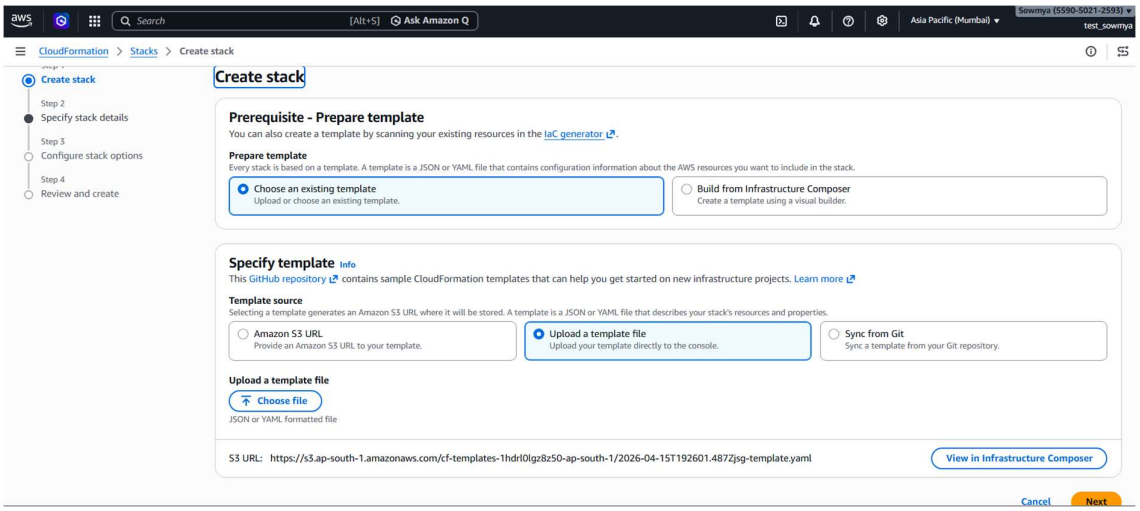

We have successfully created the bucket using template. You can also go to the default bucket created and open the template and can download too.



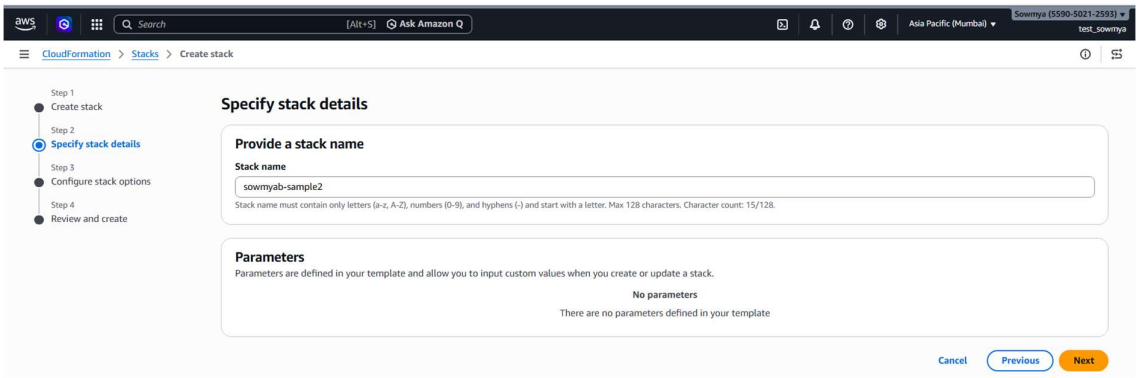

Another Method

**Step 1**: Let's try to create template with an existing template, Here I'm writing a template with the reference from

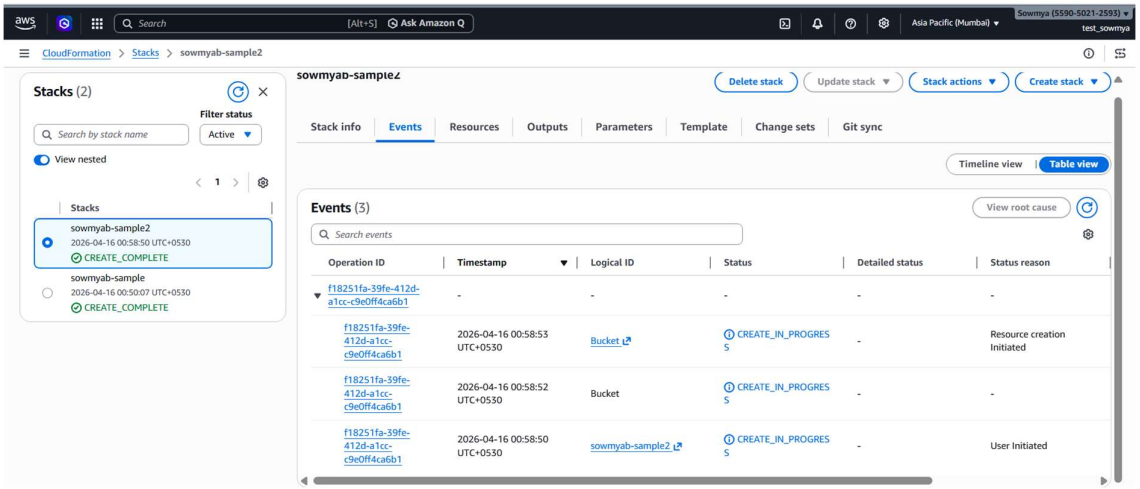
https://docs.aws.amazon.com/AWSCloudFormation/latest/TemplateReference/aws-resource-s3-bucket.html and uploaded in the console.



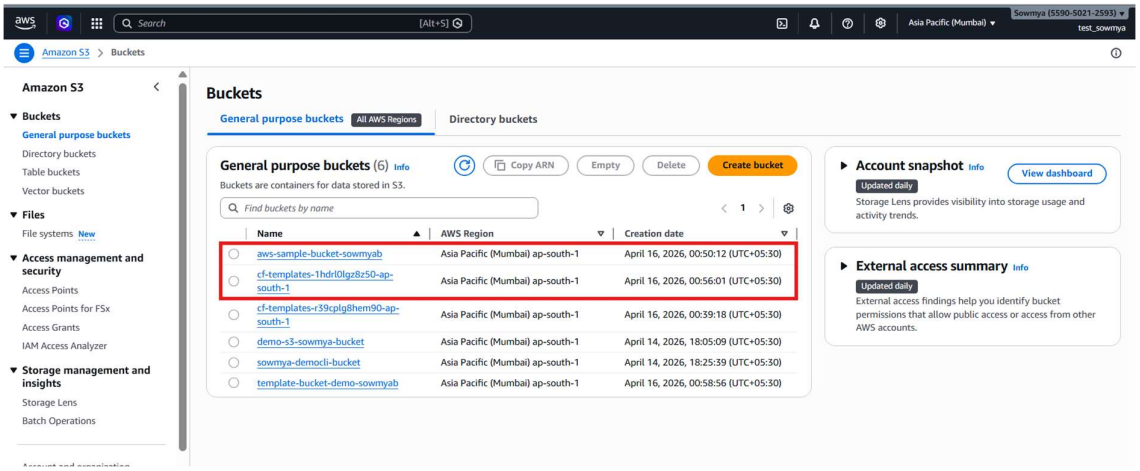

**Step 2**: Click on next, give the stack name and review everything then click on submit.



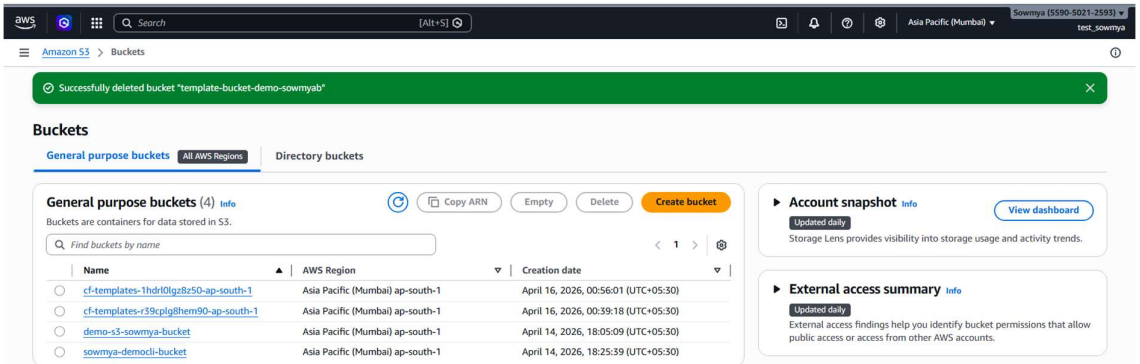

**Step 3** : You will be able to see two stack you have created in the console



**Step 4** : Search S3, go to buckets->general purpose buckets, you will be able to see the buckets and the templates.

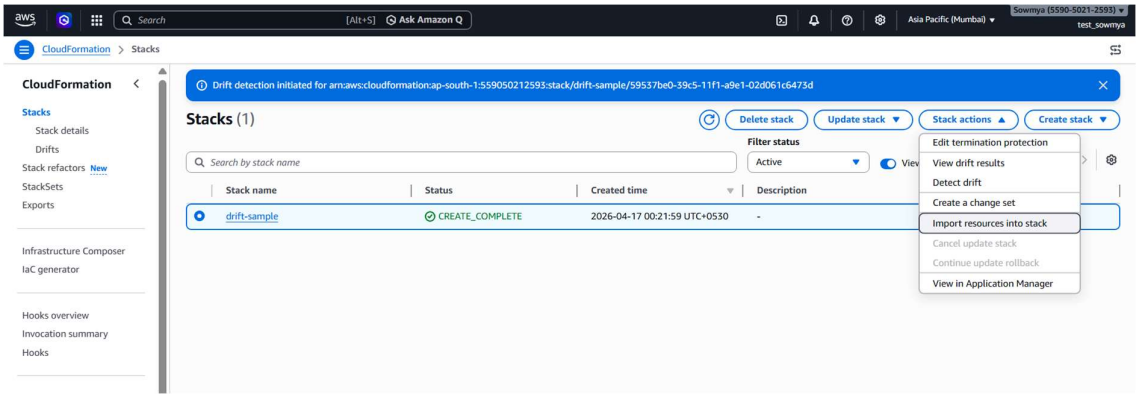



**Step 5**: You can also delete the buckets, yet the templates will remain.



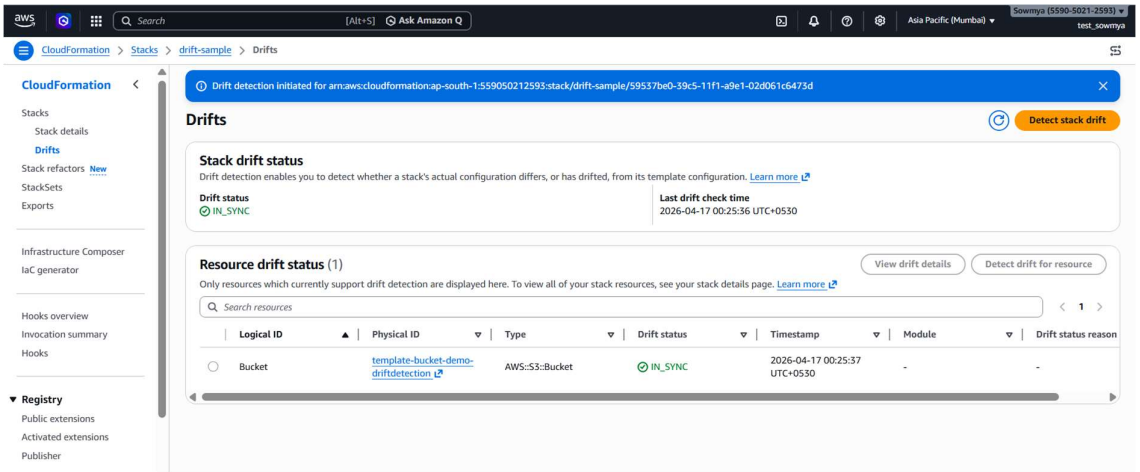

## Drift Detection

**Step 1**: Search for Cloud Formation and then select the stack name. In the right upper corner, you have stack actions in that select detect drift.

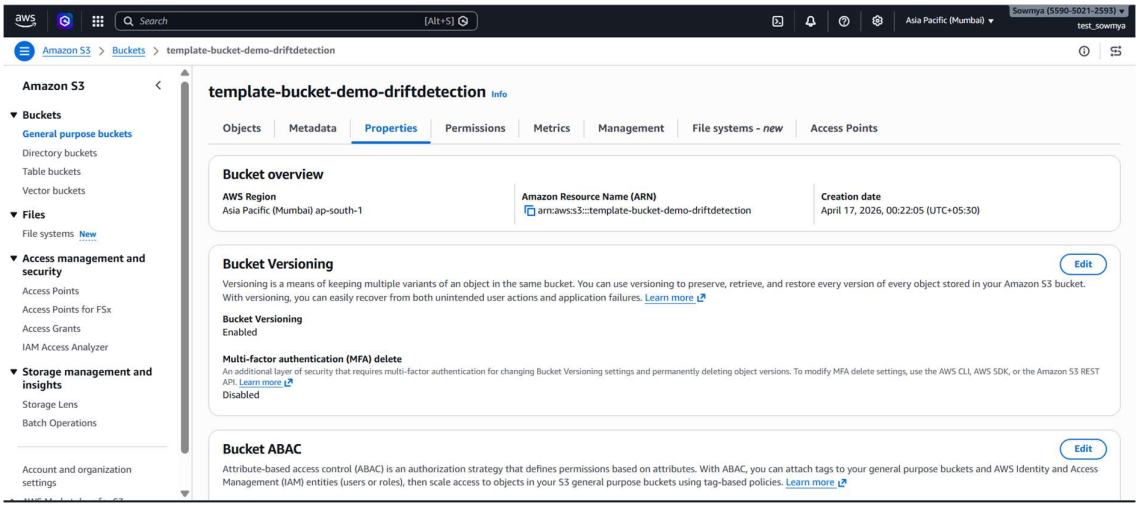


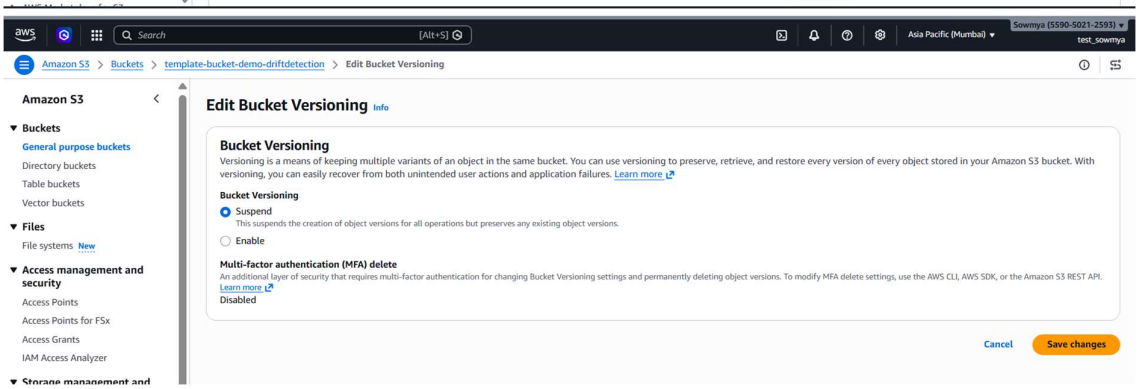

**Step 2**: In the left side panel, select drifts option. stack drift status will be displayed.

Here in the image you can see everything looks good as it is in sync



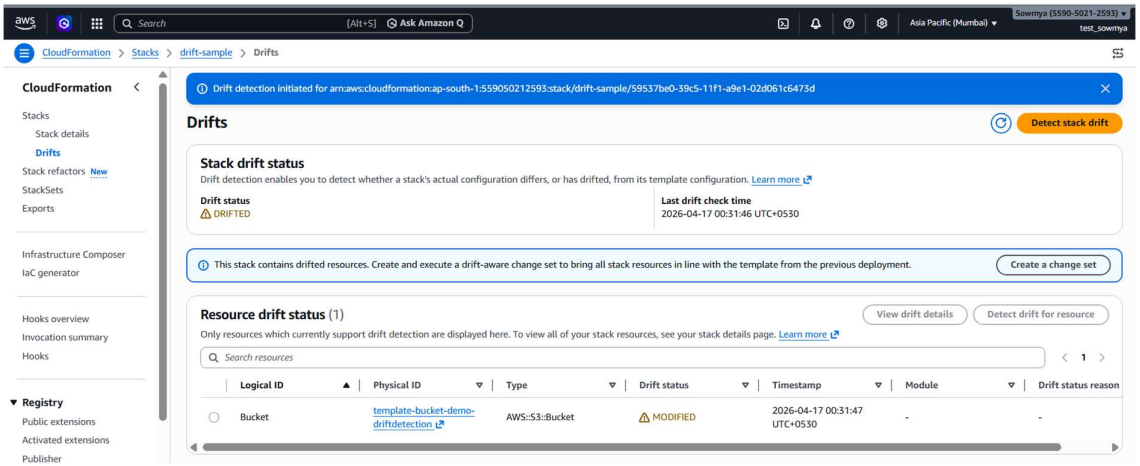

**Step 3**: Search for s3, select the bucket created->properties. As the Bucket versioning is enabled here, click on edit and change the status to suspend and click on save changes.

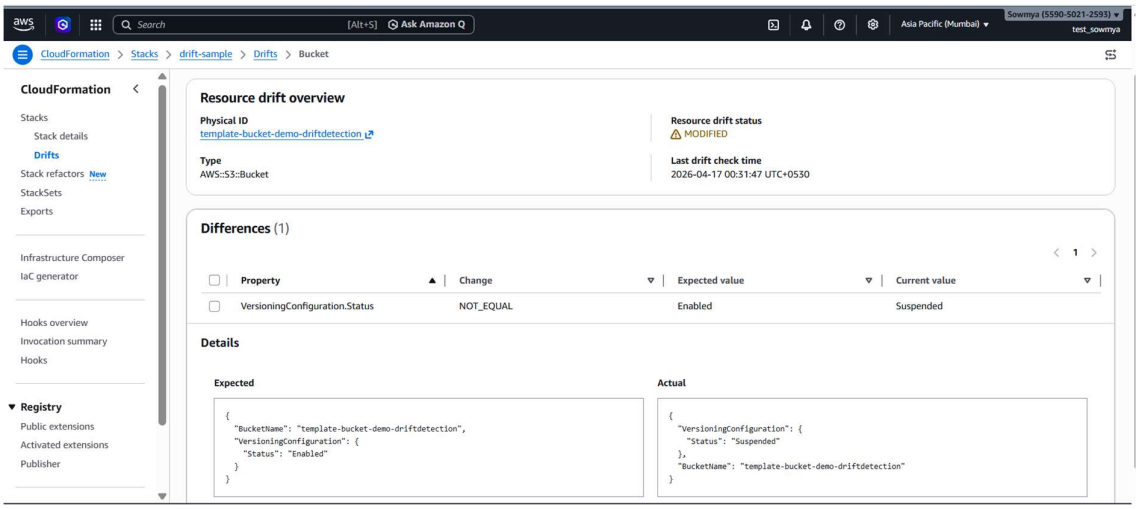



**Step 4**: Now the drift detection policy will show this change that has happened.

Go to stack->select the stack->stack actions->detect drift->drifts




**Step 5**: select the bucket and click on view drift details. You will be able to see clearly the change that has happened.

The comparison clearly highlights the difference between the **expected configuration** defined in the CloudFormation template and the **actual configuration** in the AWS environment. This mismatch results in the resource being marked as **DRIFTED**.